Mittuniversitetet
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Försättsblad tentamen / Examination cover

2707183

Anonymitetskod / Anonymous code		
S - 0004 - SYH		
Kurskod / Course code: DT081A	Provkod/ Test code: T101	Tentamensdatum / Examination date: 2024-10-29
Kursnamn / Course name: Datateknik AV, Signal- och bildbehandling		
Provnamn / Test name: Tentamen		

Skriv din anonymitetskod på varje inlämnat papper
Write your anonymous code on each sheet submitted

Sätt ett kryss (x) för varje inlämnad uppgift
Use an x to indicate which questions has been submitted

Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes	Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes
1	X	2,5		16			
2	X	1		17			
3	X	3		18			
4	X	2		19			
5	X	1		20			
6	X	2,5		21			
7	—	0		22			
8	X	1,5		23			
9	X	1		24			
10				25			
11				26			
12				27			
13				28			
14				29			
15				30			
Poängsumma / Points		14,5	Betyg / Grade: F	Lärarsign./ Teachers sign			

Fylls i av tentamensvakt / To be filled in by the invigilator

Antal lösa blad/ No. of sheets submitted	Inlämnad tentamen / Submitted exam	Leg kontroll / Control identification	Sign. tentamensvakt / Sign. invigilator
10	8	OK	T.U.

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Försättsbladet skall alltid lämnas in även om ingen uppgift behandlats
Examination cover should always be submitted even if no questions are answered

Utskriven 2024-10-28 kl. 12:37:16

Example Solutions for DT081A – Signal and Image Processing Exam, 2024-10-29

1 (1 + 1 + 1 + 1 + 1 = 5 p)

a)
The process of converting a continuous-time signal into a discrete-time signal by taking periodic samples, $x[n] = x(nT_s)$ where T_s is the sampling interval.

b)
The process of mapping a continuous range of values to a finite set of discrete levels, often in analog-to-digital conversion.

c)
The output of a system when an impulse signal is applied at its input. It characterizes the system's behavior.

d)
The steady-state response of a system to sinusoidal inputs at varying frequencies.

e)
A technique used to reduce spectral leakage in signal analysis by applying a window function to a signal before performing a Fourier transform.

2 (3 + 2 = 5 p)

a)
$y[n] = s[n] * h[n]$

$y[0] = (3)(-0.5) = -1.5$
$y[1] = (7)(-0.5) + (3)(0) = -3.5$
$y[2] = (10)(-0.5) + (7)(0) + (3)(0.5) = -5 + 1.5 = -3.5$
$y[3] = (3)(-0.5) + (10)(0) + (7)(0.5) = -1.5 + 3.5 = 2$
...

$y[n] = \{-1.5, -3.5, -3.5, 2, 7.5, 1.5, -2.5\}$

b) This filter approximates a differentiation operation, emphasizing changes in the signal, which is characteristic of a high-pass filter.

3 (1 + 2 = 3 p)

a) The system diagram shows both a feed-forward and feedback branch. The latter implies that the system will produce an infinite impulse response IIR.

b)
$y(n) = x(n) + 2x(n-1) + x(n-3) - 3y(n-1) + 2y(n-3)$

4 (2 + 3 = 5 p)

a)
Extend spectrum plot to at least $-2\pi < \Omega < 2\pi$, where the continous spectrum is mirrored on the y-axis and replicated over the whole Ω axis, with a periodicity of 2π.

b)
$f_s \geq 2f_{max}$
The lowest sampling frequency depends on the signal bandwidth. The figure shows that $f_{max} = \frac{3}{4}\pi\frac{f_s}{2\pi} = \frac{3}{8}f_s = \frac{150}{8} = 18.75Hz$. Hence $f_s \geq 37.5Hz$.

5 (2 + 4 = 6 p)

a) Amplitude and phase relationship:
$|Y(f)| = |H(f)||X(f)|$
$\angle Y(f) = \angle H(f) + \angle X(f)$

b) Sketch spectra

$$|Y(f)| = \begin{cases} 8f & \text{for } 0 \leq f \leq 10 \\ 80 - 4(f-10) & \text{for } 10 < f \leq 30 \\ 0 & \text{for } f < 0 \text{ or } f > 30 \end{cases}$$

and

$arg(Y(f)) = 0 \forall f$

6 (2 + 2 = 4 p)

a) Window iii) has the narrowest main lobe, which provides the best frequency separation between closely spaced frequency components allowing for better distinction.
b) Window iv) has zero outside of main lobe, which effectively eliminates spectral leakage in the frequency domain ensuring that frequency components are not contaminated by artifacts from distant frequencies.

7 (3 p)

At low frequencies the system has an amplification of 1, whereas higher frequencies are amplified more. Combined this results in high frequency amplification (not to be confused with a high pass filter), which if applied to an image would enhance edges.

8 (1 + 1 + 1 + 1 + 1 = 5 p)

- $1 \rightarrow ii$
  Only the center element is used, and reduce the input with 0.5.
- $2 \rightarrow iv$
  A vertical filter emphasized vertical edges only.
- $3 \rightarrow v$
  Edge detecting kernel with elements summing to 0, only shows edges.
- $4 \rightarrow iii$
  Avering kernel smooths the image as a low-pass filter.
- $5 \rightarrow i$
  Edge detecting kernel summed with a unit kernel translates kernel to one with center element = 5, with elements summing to 1.

9 (6 p)

$x(i,j)$ has several non-zero values, that each contribute to the $y(i,j)$ through convolution with $h(i,j)$. From examining the isolated scaled impulse responses we can determine that $h(i,j)$ has a cross-shaped pattern. By subtracting one scaled impulse reponse from another we can identify the impulse response as.

$$h(i,j) = \begin{bmatrix} 0 & 1/8 & 0 \\ 1/8 & 1/2 & 1/8 \\ 0 & 1/8 & 0 \end{bmatrix}$$

Lärarens anteckning / Teachers note:

Question :1 :-

a) Sampling : Sampling is the way to Convert Continuous Signal into desdrete Signal. Its really important step in signal processing because its really helps to analyze it in cligital way.
For example : if we Convert the signal in frequer domain we can have its magnitude spectrum and phase spectrum thats helps in analyzing.
Amplitude : $\sqrt{a+b}$
phase : $\tan^{-1}(\frac{b}{a})$ 0,5

b) Quantization :
Quantization is the procers of representation of Signal (Continous) into cligital (finite values). It helps us to represent the signal values in a finite domain to stucly about signal's properties and behaviours. 0,5

c) Impulse response :
The response to an impulse Input. It means it based on the short part of Input.

For example : If we have an input at some time 't' then we can have the impulse response that can have the same properties as the whole system's response. 1

if we will have Input $x(t)$ then it will have the impulse responce $y(t)$.

$x(t) \longrightarrow y(t)$

Lärarens anteckning / Teachers note:

d) Frequency response :

As we represent the signal In frequency domain to analyse its behaviour we can have the response in frequency domain. This type of response will describe how the system will behave on some frequency with respect to its amplitude and phase.

$$H(e^{j\Omega}) = |H(e^{j\Omega})| e^{j\arg(H(e^{j\Omega}))}$$

0

e) Windowing :

Windowing is the way to minimize spectral Leakage while representing the signal. Some small unwanted frequencies disturbs the system and these frequencies are removed or minimized from the signal.

Windowing can be performed by:

i) Rectangular ii) Hann iii) Hamming .etc

0,5

2,5

Lärarens anteckning / Teachers note:

Question 2 :-

$s(n) = \{3, 7, 10, 3, -5\}$

a) Convolve using Impulse response :

$h(n) = \{-0.5, 0, 0.5\}$

The general formulation for Convolution is

$$s(x*h)n = \sum_{K=-\infty}^{\infty} x(K)h(n-K)$$

Now we can subtitute the values :

for $n=0$:-

$s(0) = x(0)h(0) +$

$s(0) = 1.5$

for $n=1$:-

$s(1) = x(0)h(1-0) + x(1)h(0) = 3\times 0 + 7\times(-0.5)$

$s(1) = -3.5$

for $n=2$:-

$s(2) = x(0)h(2-0) + x(1)h(2-1) + x(2)h(2-2)$

$= 3\times(0.5) + (7\times 0) + (10\times(-0.5)) = 1.5 + 0 + (-5)$

$s(2) = -3.5$

for $n=3$:-

$s(3) = x(0)h(3-0) + x(1)h(3-1) + x(2)h(3-2) + x(3)h(3-3)$

$= 3\times 0.5 + 7\times 0.5 + 10\times 0 + 3\times(-0.5)$

$s(3) = 2$

for $n=4$:

$s(4) = x(0)h(4-0) + x(1)h(4-1) + x(2)h(4-2) + x(3)h(4-3)$
$x(4)h(4-4)$

$= 0 + 0 + 10\times 0.5 + 3\times(-0.5) + (-5)\times(-0.5)$

$s(4) = 5 + (-1.5) + 2.5 = 6$

We subtitute the value (assumed) '0' for out of scale.

We have calculated :

$s(0) = 1.5$, $s(1) = -3.5$, $s(2) = -3.5$, $s(3) = 2$ and $s(4) = 6$

- We assumed that by convolving the signal with this filter we got a signal with fluctuation in it from '1.5' to negative values and then positive values till '6'

1

Lärarens anteckning / Teachers note:

Question : 2

b) Identify operations:

Orignal Signal:

$S_o(n) = \{3, 7, 10, 3, -5\}$

Convolved:

$S_c(n) = \{1.5, -3.5, -3.5, 2, 6\}$

By Considering the filtered signal we can see that low pass filter is applied, The frequency does not more than 6 is in the system.

0

1

Lärarens anteckning / Teachers note:

Question 3:-

a) System behaviour:

This system is IIR. As we can see in the figure it has dependency not only on present and past input also it depends on output (feedback).

1

b) Difference Equation:

$$y(n) = x(n) + 2x(n-1) + x(n-3) - 3y(n-1) + 2y(n-3)$$

⇒ $x(n), x(n-1), x(n-3)$ are present and past Inputs
⇒ $y(n)$ is output
⇒ $y(n-1), y(n-3)$ are perivous output values (feedback)

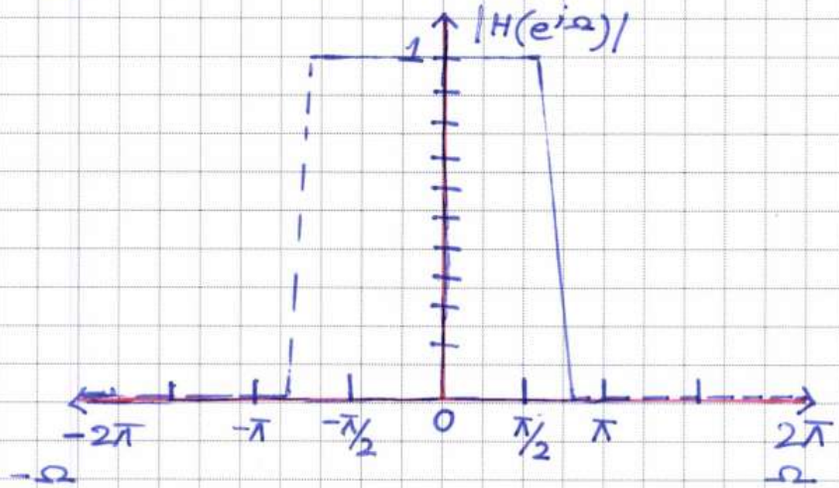
2

3

Lärarens anteckning / Teachers note:

Question :4 :-

a) $f_s = 50KHz$



If we draw the amplitude spectrum in normalized frequency range : like it is extending towards -ve side (-2π) it does not effect the amplitud because on negative side we can have the same amplitude.

$$|H(e^{j\Omega})| = \sqrt{Re(H(e^{j\Omega}))^2 + Im(H(e^{j\Omega}))^2}$$

1

b) Lowest Frequency :

- Normlized $f = 2\pi \frac{f}{f_s}$

According to Nyquist theorem :

$$f_s \geq 2f_{max}$$

if the Signal has frequency 50KHz and it is sampled on it without alising

And if the signal has sample frequency of 50KHz then the max frequency should be $f_{max} \leq \frac{f_s}{2} \Rightarrow \frac{5000}{2} = 2500\ Hz$

1

or equal

Lower than 2500Hz we can have frequecy in Signal.

- Yes it is Possibe to reconstruct it exactly.

2

Lärarens anteckning / Teachers note:

Question :5 :-

a) If $H(f)$ is our LTI system it should be

Linear : $ax_1 + bx_2 = ay + by_2$

and time Invarient : $x(t-t_0) = y(t-t_0)$.

And the system output will be :

$$Y(f) = H(f)X(f)$$

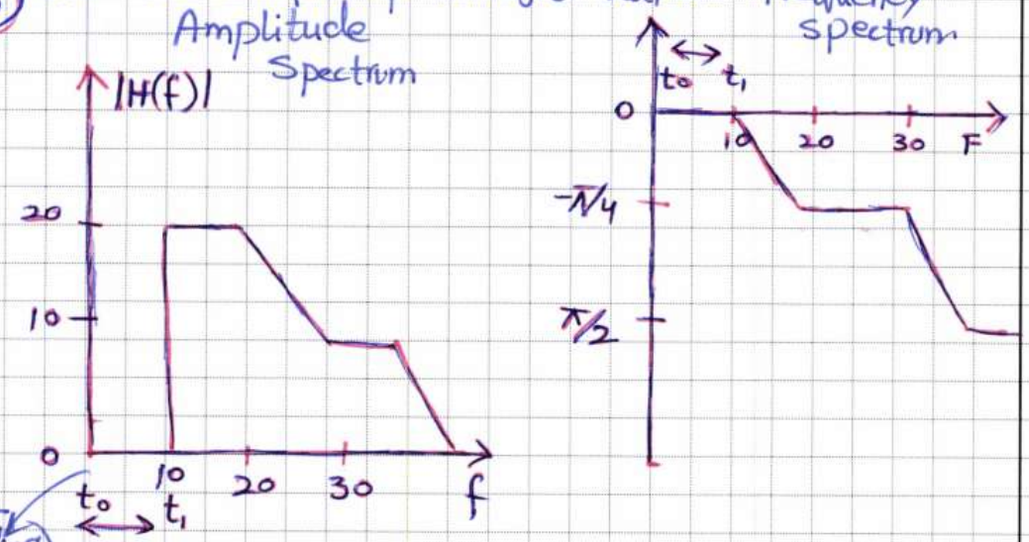
if some filter is applied to system-input then we will have the output.

$H(f)$ Consist of $|H(f)|.e^{arg(H(f))}$

$|H(f)|$ is magnitude and $e^{arg(H(f))}$ is Phase.

As it is Linear system the output will be the same as input but there can be modification and magnitude and phase values. 1

b) Its the example of delayed output. Frequency spectrum



As it is LTI the output will be the same also it can if the input have delay output will be delayed. But it will not change the output.

0/1

Lärarens anteckning / Teachers note:

Question :6 :-

a) As we can see it clearly in frequecy spectra (iii) the window function achieve best frequecy seperation.
Because In (iii) we are having more frequencies than others.
Due to more oscilations in the signal in (iii) we can say that this windowing techmique seperate more frequencies than others. 1

b) As the main focus of windowing is to remove spectral Leakage from the main frequency.
We can see in spectra (iv) the extra unwanted frequencies are removed after appling windowing.

It does not have Leakage in it and it is smooth from edges. 1,5/2,5

Lärarens anteckning / Teachers note:

Question : 8 :-

1 → (iii) 0

2 → (ii) 0

3 → (iv) 0

4 → (i) 0

5 → (v) 0

Filter results :

i) It is sharpen because its all indexes are up 0,5

ii) It is darken and noisey because luminane is out from this. 0

iii) It is blurred because higher frequency are filtered only lower f is present 0,5

(iv) higher frequency is present sharp edges shown. 0,5

v) all the higher frequencies are present high pass filtered. 0

1,5

Lärarens anteckning / Teachers note:

Question: 9 :-

As in the matrics x(i,j) we have first 3x3 block :

0	0	0
0	112	0
0	0	0

To apply 3x3 point spread function we will take the central point like (112) and multiply with each index of kernal.

Also if we will have some value in other index's in input will Add with the multplication result

For Example if we have the kernal 3x3 :-

$k(i,j) =$

a	b	c
d	e	f
g	h	i

Then for block-1 of Input :-

For block-1 :

Sample output = $y_s(i,j) =$

112xa	112xb	112xc
112xd	112xe	112xf
112xg	112xh	112xi

PSF of X(3,2) :

If we Consider the x(i,j) we can assume '64' on index x(3,2) will become

64xa	64xa	(64xa)+224
64xa	64xa	64xa
64xa	64xa	(64xa)+176

just to describe multiplication and sum in PSF

10